

COMPUTER VISION – ASSIGNMENT REPORT

Complex Computing Problem

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Introduction

This project implements a real-time human activity recognition system using MediaPipe Pose Estimation and OpenCV. The system analyses a video frame-by-frame, extracts lower-body joint coordinates, computes knee angles, and classifies the activity as Walking, Descending, or Transition using a set of rule-based conditions. No machine learning training is required.

1. Angle Computation Approach

Six MediaPipe landmarks are extracted per frame — left/right hip (23, 24), knee (25, 26), and ankle (27, 28). The knee angle at joint B, with A as the hip and C as the ankle, is calculated using:

$$\text{angle} = |\arctan2(Cy-By, Cx-Bx) - \arctan2(Ay-By, Ax-Bx)| \times (180/\pi)$$

If the result exceeds 180° , it is subtracted from 360° to keep the angle within $[0^\circ, 180^\circ]$. This angle directly reflects the degree of knee bend — a fully extended leg gives $\sim 180^\circ$, while a deep crouch drops the angle well below 90° . Tracking this in real time allows the system to distinguish between activity states without any trained model.

2. Rule-Based Classification Logic

Two signals drive classification in each frame: the vertical hip displacement and the average knee angle. The hip centre y-position is computed by averaging the left and right hip y-coordinates. The change from the previous frame gives the movement value.

- Descending — if movement > 0.003 (hip drops noticeably downward in image coordinates).
- Walking — if the descending condition is false and average knee angle $> 150^\circ$ (legs are extended).
- Transition — if neither condition is met (e.g. pausing, turning, or mid-step bend).

Both left and right knee angles are averaged to reduce sensitivity to asymmetric poses or partial occlusion of one leg.

3. Accuracy Results

The system was tested on an outdoor video of a person walking along a raised brick ledge and stepping down. Walking was correctly identified across most frames where the legs were extended (knee angles 160° – 177°). The Descending label triggered accurately as the person stepped off the ledge with a visible hip drop. Transition appeared correctly at movement boundaries.

From a visual frame-by-frame review, overall detection accuracy is estimated at approximately 80–85%. The main errors occurred near pose transitions and in the final crouching frames, where deep bending caused the landmark placement to lose some accuracy.

4. Observations

- Pose tracking was smooth and stable throughout the video with no visible flickering in the skeleton overlay.
- Outdoor daylight conditions provided clear subject–background contrast, which helped MediaPipe maintain consistent landmark detection.
- In the final crouch frames, knee angles dropped to 17° – 39° , and landmark accuracy decreased due to extreme body bend and partial self-occlusion.
- The system handles only a single subject per frame. Multi-person scenarios would require additional handling.

5. Demo Screenshots

Three frames from the processed output video are shown below, demonstrating the activity label, knee angles, and pose skeleton overlay in different states.



Fig 1 — Descending (136°/169°)



Fig 2 — Walking (171°/177°)



Fig 3 — Transition (39°/17°)

Conclusion

The project successfully demonstrates that real-time human activity detection is achievable using pose estimation and simple geometric rules. Knee angle tracking combined with hip movement detection produced accurate classifications for Walking, Descending, and Transition states, with an estimated accuracy of 80–85% on the test video. The approach is lightweight, interpretable, and requires no training data, making it a practical baseline for more advanced recognition systems.

Downloadable video link:

<https://www.pexels.com/video/a-man-doing-a-front-flip-from-a-wall-6002033/>